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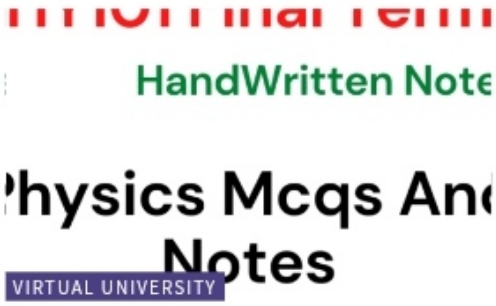
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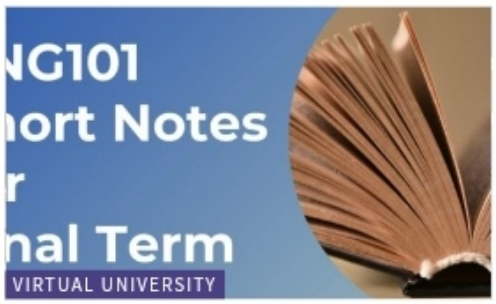

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Module 150

DATABASE SYSTEM

Database :

It is the collection of data that is multidimensional and internally linked between its entries.
It is the shared collection of data that can be retrieved, and search quickly as when we needed

Flat - File System :

Traditional file system is called flat-file system.
It is a one-dimensional storage system it means flat-file system present a information in a single point of view.

Advantages Of Database

1. Database is organized collection of records.
2. Database system have become the vast technology that supports many of the more popular sites on the world wide web.
3. Collection of data on Google, eBay and on Amazon, on online shop, and this collection of data is organized So, this is the work of database technology.

List of contacts, Guzzerts are the example of database technology.

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Module # 151

Role Of Schema

Data always stored in a hard disk in the form of 0 and 1 bit.
File-data system is also stored in the form of 0 and 1. Similarly, in a database system, data is also stored in the form of 0 and 1.
Database system become eficient and data perform a better performance, but it doesn't means it change the way of storing data.

Schema:

A schema is the description of the entire database structure that is used by the database software to maintain the database.
Data show in a table (tabular) form.
In a database system, data always stored in the form of table.

Subschema:

A subschema is a description of only that portion of the database according to a particular user's need.
Schema is a data structure, it is the way of storing data and it is managed by the database management system.

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Module # 152

Database Management System

Database Management System:
Set of computer programs which are used to create, update and maintain, control and administrate the database is called database management system.

OR

The actual manipulation of the database is accomplished by the database management system (DBMS).

Distributed Database:

It is the database which is not install physically in a one computer or stored but it install physically more than a one computer and it present on a more than one computer. So in this way, one computer cannot handle the data and we distributed the data and such a form of database is called distributed database.

Module # 153

Relational Database Model

In 1969, a mathematican E.F code present a model by using a set theory and binary relations and it consist on 12 to 13 principles and according to this model a relational database model come into being

Relations:

In a relational database model, data is being stored in a form of rectangular table known as relations

Tuple:

A row in a relation is called tuple. It also known as "TOO-pul" or "TUH-pul"

Attributes:

Column in a relation is called attributes. They also known as characteristics

Module # 154

Issues Of Relational Designs

When our relational database is best than the performance of our database system is also best. A poor database design can crash or destroy the system or the speed of system.

Relational Issues

* Duplication
* Redundancy

These two issues are settled by the process known as normalization

Module # 155

Relational Operators

Relational operators are the main part of relational model.

There are three relational operators

- Select operation
- Project operation
- Join operation

Module # 156 - 157 - 158

Select Operation
Project Operation
Join Operation

Select operation :

Select operation ~~search~~ retrieve the data which is present in database.
With the help of select operation, we can search the record from a table.

Project Operation :

Project operation extract the information from coloumn rather than a row. With the help of project operation we can get the record from a coloumn.

Join Operator :

Join operator is used to join the two relations into a one relation. For the conjunction of two relations we use a join operater. There are many type of join operator.

inner operator
outer operator
etc.

The Join of two relations produce a new relation

Module # 159

Object Oriented Database

Modern database system is known as object-oriented database.

Object-oriented Database :

Object-oriented database consisting of the objects that are linked to each other to reflect their relationship.

The links between objects in an object-oriented database are normally maintained by the DBMS.

Persistant :

Objects that are created and added to a database must be saved after the program that created them terminates. Such objects are said to be persistent.

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Module # 160

Maintaining Database Integrity

Integrity:

Integrity mean security. Inexpensive database management systems for personal use are relatively simple system.

Backup Copy:

When a problem does occur, the user can correct the problem or reload the data from a backup copy and user can also make modifications required to bring a copy up to date.

Database Integrity:

In order to solve these problems or in this environment, DBMS play a major role in the maintanance of database's integrity by guarding against problems.

Module # 161

The Commit / Rollback Protocol

Transaction :

The transfer of funds from one bank account to another, the cancellation of an airline reservation, registration of student in university, might involve multiple steps at the database level.

One single task which performed with a database is called a single transaction.

Commit Protocol :

When a transaction is successfully run then it is called commit protocol.

Rollback Protocol :

When an error is occured during transaction. So in a database system, DBMS have a power to reverse or rollback the all transaction. It may take some time and this is called commit and rollback Protocol.

Example :

ATM is example of transaction

Commit Point:

The point at which all the steps of transaction have been completed or recorded in the log is called commit point.

Rollback Point:

The point at which all the steps of transaction have not been completed or not recorded in the log, then the transaction will rollback.

Module # 162

Locking

Incorrect Summary Problem:

If one transaction is in the middle of transferring funds from one account to another when another transaction tries to compute total deposit in the bank

Example:

When account A want to transfer something in the account B. It means we want to transfer or deposit some amount from account A to B and at the same time, account B tries to compute deposit in bank

Lost update Problem:

If one transaction reads the account's current balance at the point when the other has just read the balance but has not yet calculated the new balance, then both transactions will base their deductions on the same initial balance.

Problem

These are the problems while transaction To solve such problem we have need a solution.

Solution:

* DBMS could force transaction to execute in their entirety.

So it mean that transaction A should complete its turns until that no other instructions can be executed on to CPU.

* Lock the particular item in the memory, no one can use it unless it is unlocked by the parent transaction.

No one can able to unlock or to execute or update our transaction until we unlock the transaction.

Type of Lock:

There are two types of lock.

- shared lock
- exclusive lock

Shared Lock :

The transaction is not going to change the data, this only going to read, then one can use shared lock, this means other transaction can view the record but can not update the value.

Exclusive Lock :

In this lock, the transaction is going to alter the item So this means that one thousand will be decremented or inclemented, in that situation no one is even allowed to see, that it must have exclusive access, meaning that it must be the only transaction with access to that data. and others will not be even be able to view because if they are going to view then there will be occured changing in transaction.

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Module # 163

Sequential Files

1 Why we study traditional file structures ?

⟹ We study traditional file structure because

* Database technology has evolved from traditional files structure.

For example;

Technologies like indexing and hashing are important tools in the construction of massive DBs.

* The concept of indexing and hashing - which was used by the file structure / traditional files structure. and this is very important concept in DBs.

Sequential Files :

A sequential file is a file that is accessed in a serial manner So data is stored in a serial manner. e.g First data is stored and the next and next and when you are going to read the data you need to come at the front and then read in a sequence.

Explanation :

In sequential file, from the beginnig to it's end all the information in a file were arranged in one row.

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Applications:

Files have a many applications. Audio files, vedio files, files containing programs, and files containing documents which all have a sequence.

- Most of the files created by personal computer users are sequential files.
- Excel file is stored as a sequential file.

Data storage In The Drives:

Data should be stored as a sequence.

- In case of magnetic tape or CD, the stored data is in a sequence.
- In case of magnetic disk storage, files would be scattered in different sectors. We can say the information of first employe will be in first sector and then next and next. So, the information is in a sequence form.

End-Of-File (EOF)

In processing a sequential file, we need to detect when end of the file is reached. Generally, we refer to the end of a file as the End-of-file EOF.

- One way to add a special record sentinel at the end of file.
- Another way is to place this information in the system directory file. (#, 10)

Module # 164

Indexed Files

Why we need indexed Files ?
When information needs to be retrieved in a sequence, sequential file is fine. However, when one wants to retrieve in unpredictable form/manner.

Explanation :

A file index is normally stored as a separate file on the same mass storage device as the indexed files. The index is usually transfer to main memory before file processing and main memory tell us which memory locate where and also tell that which record is where. Then we going to a particular locate and get a information

Module # 165

Hash Files

Hashing :

- It is a technique that provides a similar access without storing index as overhead.
- Instead of looking into index, hashing

identifies the location of the record directly from the key.

Explanation:

If we are interested to find any record in the memory. So we will directly reach the address immediately we do not need to wait

Hash System:

The data storage space is divided into several sections, called buckets, each of which is capable of holding several records directly from the key.

Hash Function:

The records are dispersed among the buckets according to an algorithm that convert key values into bucket numbers. This conversion from key values to a bucket numbers is called a hash function

Hash File VS Hash Table:

- When hashing is applied to a storage structure in mass storage, it is called hash files.
- When hashing applied to a storage structure within main memory, then we called it hash table.

Module # 166

Hash Files Example

Hash Function:

- Hash function is the key factor to store and retrieve data.
- Hash function can be made using variety of ways.

One simple method is remainder method.

How we will find a function that will tell us that this record should go in that particular bucket or not. So

For example:

We have a ten buckets so we need to find record no six 6. So, this record no six going in which buckets available to us. This kind of thing is called hash function.

Remainder Method Example

$$10 \overline{)12} \quad 1 \quad \text{remainder } 2$$

$$10 \overline{)22} \quad 2 \quad \text{remainder } 2$$

$$10 \overline{)2} \quad 0 \quad \text{remainder } 2$$

We can see that record no 12, 22 and 2 all are putted into bucket 2.

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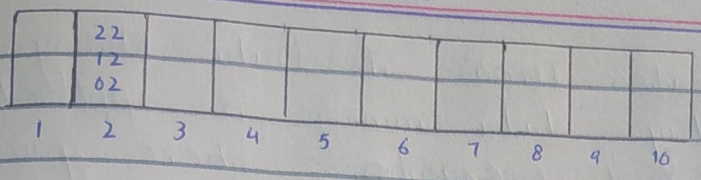

Clustering :

Goal of Hashing

The records being stored should be distributed evenly among the buckets.

Clustering

Disproportionate number of keys happen to hash to the same bucket.

So this means majority or most of the records are coming at a one particular buckets.

As we have seen that all of the remainder are 2 then second number bucket will be over loaded and all of the other buckets will be empty.

This concept is known as clustering.

Module # 167

Data Mining

A rapidly expanding subject that is closely associated with database technology is data mining., which consist of techniques for discovering patterns in collections of data.

Application Areas of Data Mining:

Today, data mining is applied in almost everywhere, where you can find the data. Data mining is one of the core thing in data science. This have application areas are

- Marketing
- Inventory management
- Quality control
- Loan risk management
- Fraud detection
- Investment analysis of data
- Identifying the functions of

Data Warehouses:

Data mining is practiced on static data collection called data warehouses rather than "online" operational databasis that are subject to frequent updates. These warehouses are often "snapshots" of databases or collection of databases.

Common Forms of DM:

There are two common forms of DM/data mining.

Class description

Class discrimination.

Class description:

This deal with identifying properties that characterize a

given group of data items.
example:
To identify characteristics of people who buy small economical vehicles
Class descrimination:
This identify the properties that deal with two groups.
Example:
To find properties that distinguish customers who shop for used cars from those who shop for new one.

Module # 168

Data Mining Examples And Implications

1. Cluster analysis:
It seek to discover classes, cluster analysis tries to find properties of data items that lead to the discovery of grouping.
Example:
In analyzing information about People's ages who have viewed a particular motion picture, cluster analysis might find that the customer base breaks down into two age groups.
a-0 to 4 age group
a-25 to 40 age group

2- Association Analysis

This involves a looking for links between data groups.

example :

It is association analysis that might reveal that customers who buy potato chips also buy cold drinks.

3- Outliers Analysis :

It tries to identify data entries that do not comply with the norm.

Outlier analysis can be used to identify errors in data collection.

It identify credit card theft by detecting sudden deviations from a customer's normal purchase pattern.

4- Sequential Pattern Analysis :

It tries to identify patterns of behaviour over time.

example :

Sequential pattern analysis might reveal trends in environmental systems such as climate conditions.

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Module # 169

Social Impact Of Database Technology

Database Technology has affected our lives in many aspects.

Automated library systems place a patron's reading habits within easy reach.

When you reading online library which is safe that can be used in variety of ways. So this is a social impact of database technology.

Internet search engines keep record of their clients requests.

When you request in the search engine they are recording all of your query. Then afterwards they utilize your information in variety of ways.

Data Collection:

It was a volunteer who fill the survey or government agencies make a regulations when you need to give information.

So, all of those things were only focus on volunteer surveys or the law made by government.

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- However, with database technology, data is being automatically retrieved, and saved and even the person whose data have saved into the DB he are does't know.
  e.g Grocery store ask customers to fill forms to offer discounts. So, this is another way of storing data.

Automatic Data Collection:

- Credit company records the purchasing practices of customers and they do not tell. Simply they are saving practices
- Websites record who visits the site.
- Social activist who record the liscence plate number on cars parked in a targeted institutions parking lot.

Data is currency:

- It help the targeted marketing
- Data is linked
- Subcription forms for body-building magzines can be mailed to those who have recently purchased exercise equipment.
- Similarly, dog obedience magazines can be targeted toward those who have recently purchased dog food.

Data Protection:

Here, we learn that the data should be protected

Several approaches to protecting a society from abusive use of data.

Legal remedies.

Passing a law is not enough. It does not stop the action, it merely makes it illegal.

Sharing of contact information.

So data protection can be handled by using one of another way that is known as Public opinion.

Public Opinion:

In the early 1990s, it was public opinion that ultimately stopped major credit bureaus from selling mailing lists for marketing purpose and it really worked.

Google discontinued its Google Buzz social networking tool in 2011 after public criticizm on automatic sharing of contact information.

Module # 170

Artificial Intelligence Introduction And Vision

Artificial intelligence is the field of computer science that seeks to build autonomous machines, machines that can carry out complex task without human intervention.

Applications of Artificial Intelligence:

Applications of artificial intelligence are

- Robot
- Self-Driving car

Medical Robots

These robot can allevate patients even 40 times a day.
If a patient want to allevate from wheel chair to bed or chair to washroom, then medical Robot helped him.

Peeper Robot:

Peppes Robot works as receptionist in Belgian hospital.
This robot actioned as a receptionist.
We can ask different questions from these robots. It can make appointment for us and performed different actions.

There is another very important report which is taking blood from your veins So it is used in

- Image analysis to find veins
- Confirmation by ultrasound
- Accurate then human

Explanation:

A person who going to hospital And where your blood has taken from your vein and sometime

he not find your veins and putting needle again and again and you feel pain. So such a robot has been built which you give your arm and then it performs image analysis to find multiple veins ways. So, in your hand it can see all of the veins and then it performed a ultrasound to see that which vein has more blood pressure then if we insult a needle in veins it gives a accurate amount of blood.

Applications

- TUG robot can carry 400 KG of medication from one place to another. To carry 400 kg for a Human is very difficult.
- Millimeter sized Microbot Delivers Drug through Bloodstreem at targeted area. whereever you want.

Others Applications

Cybersecurity Defence
Recruiting automation
Predicting Vulnerability
Market Prediction
Accelerated Reading
Reduced energy used and cost.

These all are the application of artificial intelligence. some of them are example of robot system.

Artificial Intelligence As Field:

Merges with other subjects such as Psychology, neurology, mathematics, linguistics and electrical and mechanical engineering targeted area.

Module # 171

Intelligent Agents

Agent:

Much of the research in a artificial intelligence can be categorized in terms of an agents behaviour. An agent is a device that respond to stimuli from it's environment. It is natural to envision an agent as an individual machine such as robot.

Agent Example

Agent may take other forms such as an autonomous aeroplane, a character in an interactive vedio game, or a process communicating with other processes over the internet

Now, How an agent acquire the data

Data Acquisition Agent

Most agents have sensors by which they recieve data from their environment and actuators (means their belonging)

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by which they can affect their environment.

Sensor Example

- Microphone

Agents use microphone to analyze, to record.

- Cameras
- Range sensors
- Air or soil sampling device

Actuators Example

- Wheels
- Legs
- Wings
- Grippers
- Speech synthesizers

All those kind are called actuators from which they can make affect, from which they can respond or behave.

Artificial Intelligence Research

- AI research can be characterized in the context of building agents that behave intelligently, meaning that the action of the agents actuator's must be rational responses to the data recieved through it's sensor.

So this means most of the time, the AI research based on agents and agents.

- Level of responses

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Levels of Responses:

- Simplest response is a reflex action. which is already program, whenever we want to do this thing and that. This is kind of reflex action or preplaned thing which has been built in the Robot.
- Intelligent Action - Adjust actions according to environment knowledge so, Agent can be made in a way that is see in this situation what I do

Example:

The Process of throwing a baseball is largely a reflex action but determining how and where to throw the ball - require knowledge of the current environment In some cases an agent responses improve over time as the agent learn from the environment. This could be taken from the form of developing procedral knowledge. (learning how) or storing declarative knowledge (learning what).

Module # 172

Research Methodologies

AI Domain

AI domain can be persued along two path.

- Engineering Track
- Theoratical Track